

S2S archive comparison — NOAA SFS beta1 vs CESM2 S2SHINDCASTS

Compiled from live S3 / Glade contents

September 11, 2026

Overview

Three co-existing archives feed the current S2S / blocking / kf_pan analyses:

- **SFS reforecast** — full ensemble × init × lead structure kept intact; right feed for lead-day and probabilistic analyses.
- **SFS MDTF** — ensemble and lead axes discarded to produce long chained daily records that behave like a free-running run; the natural feed for MDTF PODs, WK spectra, and long-record climatologies.
- **CESM2 S2SHINDCASTS** — CESM2/CAM6 subseasonal hindcast archive on Glade (weekly cadence, 22 members), split across two companion NetCDF trees: daily/ (CAM h2 output) and Z3/ (geopotential height on pressure levels).

Format comparison

Property	SFS reforecast/{MM}/atm_daily.zarr	SFS MDTF/daily_mem000_{MM}.zarr	CESM2 S2SHINDCASTS
Purpose	Probabilistic S2S hindcast — lead-day scoring, ensemble spread	MDTF POD framework — feed as if free-running run	Weekly CESM2 S2S hindcast — lead-day skill, blocking, S2S predictability
Path root	s3://noaa-oar-sfsdev-pds/experiments/beta1/reforecast/	s3://noaa-oar-sfsdev-pds/experiments/beta1/MDTF/	/glade/campaign/cesm/development/cross-wg/S2S/CESM2/S2SHINDCASTS/
Sub-trees	One store per init month: {MM}/atm_daily.zarr (+ ocn/ice, monthly, clim)	One store per init month: daily_mem000_{MM}.zarr	Two co-registered trees: daily/*.nc (CAM h2 daily) + Z3/{YYYY}/{MM}/*.nc
Format	Consolidated Zarr v2, blosc(lz4), anon-readable S3	Consolidated Zarr v2, blosc(lz4), anon-readable S3	NetCDF-4, local POSIX (one file per start × member)
Init months available	03, 04, 05, 06, 07, 08, 11 (no 09 daily)	03, 04, 05, 06, 07, 08, 09 , 11	All 12 months
Init cadence	Monthly — 1st of MM only	Monthly (already chained)	Weekly — 7-day delta, ~52 starts/year (98 % Mondays)
Starts per year, per member	7 (one per available month)	folded into single time	52-53
Year coverage	1991 → 2025 (35 y)	1991 → 2025 chained (35 y in one series)	1999 → 2023 (25 y)

Property	SFS reforecast/{MM}/atm_daily.zarr	SFS MDTF/daily_mem000_{MM}.zarr	CESM2 S2SHINDCASTS
Total unique starts	245 per member (7 × 35)	(none exposed)	1,316
Members	11 ('000'..'010' strings)	1 (mem000 baked in)	up to 22 (m00..m21, some years >22)
Lead length	47 days	folded into time	46 days
Dims (analysis vars)	(member=11, init=35, lead=47, lat=181, lon=360)	(time=12784, lat=181, lon=360)	daily/: (time=46, lat=192, lon=288), plus (ilev=33, zlon=1) for zonal-mean TEM vars. Z3/: (time=46, lev_p=14, lat=192, lon=288)
init / start encoding	int64 days since 1991-{MM}-01, one per year	none	Encoded in filename : Z3_cesm2cam6v2_{DD}{mon}{YYYY}00z_..._m{MM}.nc also in time[0]
lead encoding	int64 days, 0..46	none	time is verification-date; lead = day index 1..46
time encoding	none (init × lead)	float64 seconds since 1970-01-01, verification-time, chained 1991-03-01 → ~2026-02-28	float64 CF datetime, calendar=noleap, 46 daily verification dates per file
Grid	1° global regular, lat=181 (−90..90 asc), lon=360 (0..359°E)	same	CAM FV ~ 0.9° × 1.25° , lat=192 (−90..90 asc), lon=288 (0..358.75°E)
Fill / missing	9.999e+20 (float32)	same	CF_FillValue, per-var
Chunking (HGT/Z500 var)	[1, 1, 47, 181, 360] — one (member, init) at a time	[10, 181, 360] — 10 days at a time	One file per (start × member) — no internal chunking
Total files / stores	7 stores (one per MM)	8 stores (one per MM)	16,746 daily/*.nc files; ~similar count under Z3/
Local cache format	{cache_dir}/z500_hindcast_m{MM:02d}_{YYYY}.nc (built by build_z500_cache_from_sfs)	{cache_dir}/z500_hindcast_m{MM:02d}_{YYYY}.nc (built by build_z500_cache_from_sfs)	{cache_dir}/z500_hindcast_m{MM:02d}_{YYYY}.nc (built by build_z500_cache), same layout, plus prect_hindcast_m{MM:02d}_{YYYY}.nc for kf_pan
Time semantics	Forecast state at (member × init × lead)	Verification-date state (single member)	Forecast state at (member × start_date × lead_day)
# variables	~80	75	27 (h2 daily on lat/lon) + 8 (zonal-mean TEM on ilev/zlon/lat) + 1 (Z3 on 14 pressure levels)
Best for	Arbitrary (member, init) slicing; lead-day pools; ensemble stats	Streaming a long daily series; MDTF PODs; WK / spectra / climatology	Weekly lead-day analyses; blocking-by-lead; PRECT kf_pan; large ensemble variance

Variable content — where do the fields overlap?

CESM daily/ (CAM h2) uses CAM/CMIP-style names, not GRIB. The table below shows the *rough* equivalences for the fields that come up in these analyses. Z3 sits in a separate archive (Z3/{YYYY}/{MM}/*.nc) and holds heights on 14 pressure levels: 5, 10, 20, 30, 50, 70, 100, 200, 300, 500, 700, 850, 925, 1000 hPa.

Category	SFS reforecast	SFS MDTF	CESM S2SHINDCASTS
Precipitation	PRATE_surface (kg m ⁻² s ⁻¹), APCP_surface (accum)	PRATE_surface only	PRECC + PRECL (m/s); PRECT = (PRECC+PRECL) × 86.4e6 mm/day
Radiation surface (down)	DSWRF_surface, DLWRF_surface	none	FSDS, FLDS (both W m ⁻²)
Radiation surface (up)	USWRF_surface, ULWRF_surface	none	not in h2
Radiation TOA	DSWRF_topofatmosphere, USWRF_topofatmosphere, ULWRF_topofatmosphere	ULWRF_topofatmosphere only	FSNT (net SW TOA), FLNT (net LW TOA)
Cloud (total)	TCDC_entireatmosphere_consideredasasinglerepresentation	none	CLDTOT
Cloud water/ice column	none	none	TGCLDLWP, TGCLDIWP (column liquid/ice)
Column moisture	none	none	TMQ (precipitable water, kg m ⁻²)
Sea-level pressure	PRMSL_meansealevel	none	PSL
Surface pressure	PRES_surface	PRES_surface	PS
Heat fluxes	LHTFL_surface, SHTFL_surface	LHTFL_surface only	QLFX (moisture flux, kg m ⁻² s ⁻¹) — no SHFLX in h2
Sea ice / SST	none	ICEC_surface	SST (K); no ice conc
Potential evap	none	PEVPR_surface	not in h2
Vertical wind	VVEL_500mb	VVEL_500mb	zonal-mean Wzm only (no 3-D field in h2)
Winds (pressure levels)	UGRD/VGRD @ 50–1000 mb (11 levels)	same	not in h2 (only surface U10, WSPDSRFAV, WSPDSRFMX; zonal-mean Uz, Vz)
Winds (10 m)	UGRD_10maboveground, VGRD_10maboveground	same	U10 (10 m wind speed only)
Heights (pressure levels)	HGT @ 50, 100, 200, 500, 700, 850, 1000 mb + surface	same	Z3 archive — 14 lev_p, includes 500 mb → Z500 = Z3.sel(lev_p=500)
Temperatures (pressure levels)	TMP @ 50, 100, 200, 250, 300, 500, 600, 700, 850, 925, 1000 mb + 2 m + surface	same	not in h2 (only TROP_T at tropopause; zonal-mean THzm)
2 m T, RH, Q	TMP_2maboveground, SPFH_2maboveground	same	QREFHT (2 m Q), RHREFHT (2 m RH), RH600 (600 mb RH)
Humidity (pressure levels)	SPFH @ 100–1000 mb	same (no 50 mb)	not in h2
Tropopause	none	none	TROP_P, TROP_T
Zonal-mean TEM diagnostics	none	none	Uzm, Vz, Wzm, THzm, UVzm, UWzm, VTHzm, WTHzm (on ilev × lat)
Snow	none	none	SNOWHICE, SNOWHLND (snow depth)
Soil moisture / temperature	SOILL, SOILW (4 layers), CISOILM_0M2mbelowground	same	not in h2
Land / topo	LAND_surface	LAND_surface	LANDFRAC, OCNFRAC, PHIS (surface geopotential)

Key CESM-specific gaps to note. The h2 daily archive is deliberately lightweight — full 3-D fields on model or pressure levels are *not* included outside of Z3 (heights). For anything requiring winds, temperature, or humidity on pressure levels (e.g. jet diagnostics, MJO composites), the SFS reforecast is the *only* one of these three archives that ships all three in a single store.

Minimal readers

```
import fsspec, xarray as xr

# 1. SFS reforecast – 5-D probabilistic slice
mp = fsspec.get_mapper(
    's3://noaa-oar-sfsdev-pds/experiments/betal/reforecast/03/atm_daily.zarr',
    anon=True)
ds_r = xr.decode_cf(xr.open_zarr(mp, consolidated=True), decode_timedelta=False)
z_r = ds_r['HGT_500mb']          # dims (member, init, lead, lat, lon)

# 2. SFS MDTF – chained verification-time series (member 0 only)
mp = fsspec.get_mapper(
    's3://noaa-oar-sfsdev-pds/experiments/betal/MDTF/daily_mem000_03.zarr',
    anon=True)
ds_m = xr.decode_cf(xr.open_zarr(mp, consolidated=True))
z_m = ds_m['HGT_500mb']        # dims (time, lat, lon)

# 3. CESM S2SHINDCASTS – one NetCDF per (start, member); use hindcast_blocking_utils
import hindcast_blocking_utils as hbu
df = hbu.list_hindcast_files(hbu.DATA_DIR, years=[2015])  # 22 members × 52 starts
z500 = hbu.load_z500_for_leadday(lead_day=10, cache_dir=hbucache_dir)
# dims (hindcast, member, lat, lon)
```

Rules of thumb

- **Lead-day + full 3-D atmosphere** → SFS reforecast (winds, T, Q on 9–11 pressure levels).
- **Lead-day + geopotential blocking** → either SFS reforecast *or* CESM (both ship Z500).
- **Weekly-cadence lead-day skill, PRECT kf_pan** → CESM (weekly starts give the 7-day cadence chaining works out of the box).
- **Long chained daily record for MDTF PODs, WK spectra, climatology** → SFS MDTF.
- **Ensemble spread** → SFS reforecast (11) or CESM (22), depending on which variables you need.

Cache layout produced by these analyses

All three ingest paths converge on the same per-(member, year) NetCDF layout so the downstream blocking / kf_pan code is source-agnostic:

- Filename : z500_hindcast_m{MM:02d}_{YYYY}.nc or prect_hindcast_m{MM:02d}_{YYYY}.nc
- Dims : (start_date, lead_day, lat, lon)
- Var : Z500 in m or PRECT in mm/day
- Builder : hindcast_blocking_utils.build_z500_cache() for CESM, hindcast_blocking_utils.build_z500_cache_from_sfs() for SFS reforecast

Point CACHE_DIR at the SFS or CESM cache and every downstream helper works unchanged.